Responsible division :	Responsible unit :	Document type :	Confidentiality status :	
PGR	MLN/ENG	Requirement specification	Internal	BOMBARDIER
Prepared : 2019-12-03	O. Di Domenico	Title : **System Functions Req Spec Drive Control SFRS V300Zefiro**	Document state : Released	
Checked : 2019-12-04	F. Brughera		**3EST000225-7909**	
Approved : 2019-12-05	M. Brusco	File name : 3EST000225-7909_Aen.docx	Revision : _A	Language : en / Pages : 1/45
		External Revision Date: External Approver:	Customer External Identity:	
		Saved: 2019-12-05 18:00	Sequence index : 1	

Drive Control-VF.CONCEPT

SFRS for Zefiro-Europe

Generated from DOORS module /Platform.-.Zefiro-Europe/01.-.Requirements Breakdown Structure.-.Zefiro-Europe/2F.-.Vehicle-Functional Integration Areas.-.Zefiro-Europe/2F_05.-.Drive and Brake Control-VF.-.Zefiro-Europe/2F_05.01.-.Drive Control-VF.-.Zefiro-Europe/Drive Control-VF.CONCEPT, baseline 3.0

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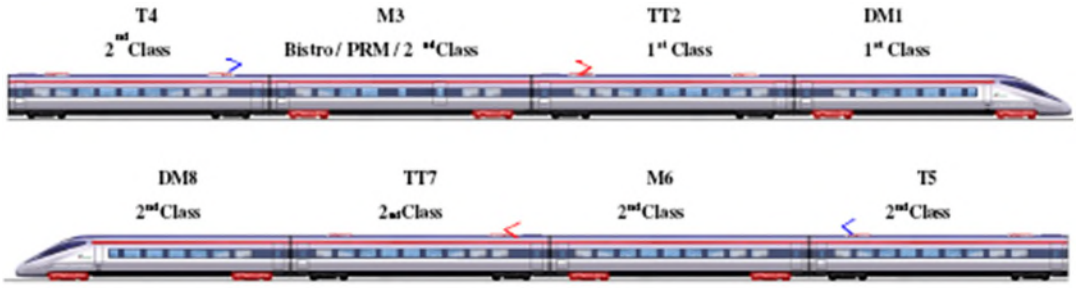
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1 Introduction

This document is a function specification for the control of the *propulsion* system for V300ZEFIRO for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end.



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

ACM	Auxiliary Converter Module
BCU	Brake Control Unit
CCU	Central Computer Unit
CCU-O	Central Computer Unit – Operation (used for vehicle control)
DC	Direct Current
DCU	Drive Control Unit
DCU/L	Drive control unit line converter
DCU/M	Drive control unit motor converter
DM	End driving motor car with driver's cab
ED	electro dynamic (ED) brake off
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
FCM	MF-DC Converter Module
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps
ICD	Interface Control Document
ID	Identifier
IDU	Intelligent Display Unit
I/O	Inputs / Outputs
LCM	Line Converter Module
M	Intermediate driving motor car
MCM	Motor Converter Module
MVB	Multifunction Vehicle Bus
N/A	Non Applicable
PCU	Propulsion Control Unit
SFRS	System Functions Requirement Specification
SW	Software
TBD	To Be Defined
TCMS	Train Control and Management System
TDS	Train Diagnostic System
TRD	Technical Requirement Document

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TRS	Technical Requirement Specification
T	Intermediate trailer car
TT	Intermediate trailer car with transformer
WSP	Wheel side protection
WTB	Wide Train Bus

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2 System Overview

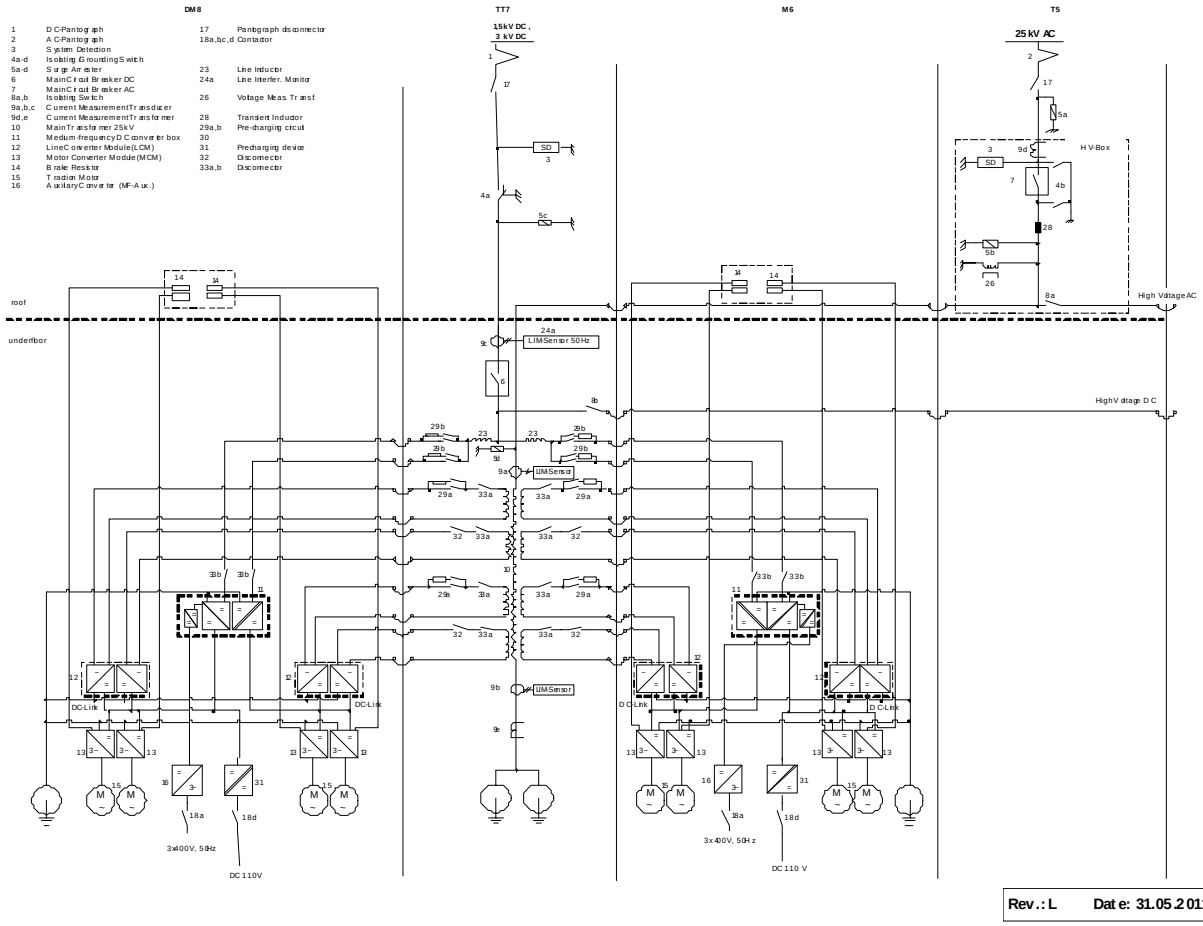
2.1 System Configuration

In each motor car (DM and M cars) there is one complete converter unit.

DM and M cars are equipped with two LCM, one ACM and 4 MCM and two FCM. They are controlled by the PCU Unit which also communicates with TCMS via MVB.

Each LCM controls two traction motors. The communication between TCMS and the propulsion system is achieved through Propulsion Control Unit (PCU). There is one PCU per LCM.

An overview of the electrical power system:



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3 Functional requirements

The traction system configuration is modular consisting of two independent high voltage systems (pantograph, main transformer) and eight independent traction converter systems. The availability of at least 50% of traction power of a train is guaranteed in any single failure case.
To assure 50% traction also in case of fire, separation of the traction systems of the two semi-trains is made possible by isolation contactors.

3.1 Driving direction vs. Motor Rotation

3.1.1 Function Design

The driving direction pushbuttons at driver's desk are specified in the SFRS for Conventional Train Control.

In each PCU, the driving direction shall be checked in order to avoid a mismatch with the rotating direction of a motor.

Detected mismatch between driving direction and motor rotation will be indicated by the PCU only as a "mechanical fault" and the respective MCM will be shut off.

3.1.1.1 Normal condition

MCM shall control that the traction motor rotating direction of each motor is corresponding to the chosen driving direction.

3.1.1.2 Failure cases

Description of failure cases

A) The wrong direction of the motor is not detected.
B) No communication between TCMS and PCU. Not possible to indicate failures to the driver. It is not likely that the motor is rotating in the wrong direction since traction will be disabled at the motors where the communication has been lost between TCMS and PCU. This implies a possible reduction of max 12,5% traction power.
C) The motor is not shut-off when rotating in the wrong direction, i.e. that the protective blocking is not working on the motor. Risk of wheel and motor damage depending on the adhesion.

3.1.1.3 Failure detection

Description of failure detection

A) The faulty direction is detected by the mechanical fault detection function in the MCM.
   No failure detection for this failure case.
B) The communication between CCU and the PCU is supervised by the CCU which will trigger an event stored in the diagnostic database defined in the TCMS Operational SFRS.
C) None

3.1.1.4 Failure handling

Description of failure handling

A) The MCM is blocked. To be maintained at next scheduled workshop visit.
B) Fault will be logged in the diagnostic database and checked at maintenance. Unscheduled workshop visit at the earliest after two days.
C) Maintenance: Major failure.

3.1.2 Requirements

ID	2F_05.01.Zefiro-Europe.TRS.184
Description	The torque direction exerted by the traction motor (both, in traction or dynamic braking) shall be monitored and compared to the driving direction requested by the driver.
Safety level	SIL 1

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ID	2F_05.01.Zefiro-Europe.TRS.186
Description	Driving: In case of discrepancy (traction motor vs. driver request direction), the torque shall immediately be canceled.
Safety level	SIL 1

The rotation direction of the motor shall be supervised and compared to the driving direction sent by MVB and trainlines. In case of discrepancy the motor torque shall be stopped.

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.207	In case of mismatch between the rotation of the motor and the demanded driving direction, the motor shall be shut off.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_05.01.Zefiro-Europe.CONCEPT.615	If a mismatch is detected between the rotation of the motor and the demanded driving direction, an event shall be generated only indicating "mechanical fault".	Derived Requirement	4S_05.-.Propulsion	EVENT	N/A

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3.2 Train Weight Correction Factor for traction

3.2.1 Function Design

The performance of the train shall be independent of its load. To achieve this performance, the traction demand shall take in account the load weigh (passengers, baggage etc.) of the train.

The train mass is measured. The train weight correction factor is directly proportional to the local car weight. The tractive reference is scaled according to the train weight correction factor.

3.2.1.1 Normal condition

In normal condition, the train weight is measured in each car by the secondary suspension measured by the Braking system in each car. The car weight will be forwarded to TCMS which will calculate the total train weight.
The train weight correction factor will be calculated by TCMS and sent to each PCU which will compensate traction demand accordingly.

3.2.1.2 Failure cases

Description of failure cases
A) Train weight is invalid.
B) Communication fault between TCMS and PCU. The train weight correction factor will not be sent to the PCU control but that is secondary since the communication fault disables the traction motors where the communication has been lost between TCMS and PCU.
C) Calculated Train weight is less than the physical train weight. This fault will imply that there is an undetected fault in the secondary suspension device.
D) Calculated Train weight is greater than the physical train weight. This will imply that there is an undetected fault in the secondary suspension device.

3.2.1.3 Failure detection

Description of failure detection
A) The secondary suspension is supervised by the Braking system.
B) The communication between CCU and the PCU is supervised by the CCU which will trigger an event stored in the diagnostic database.
C) Detected by driver due possible problems reaching maximum train speed.
D) No detection available that can be considered as valid.

3.2.1.4 Failure handling

Description of failure handling
A) The train weight correction factor will be set to 100% to allow full performance. An event informing invalid train weight will be set in the diagnostic database.
B) Fault will be logged in the diagnostic database and be checked at maintenance. Unscheduled workshop visit at the earliest after two days.
C) Checked during maintenance based on fault reports by the driver.
D) None identified. The train weight correction factor will be set to 100% to allow full performance.

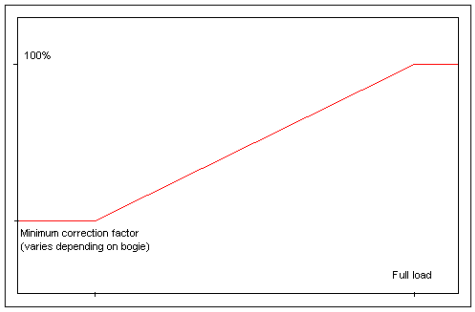
3.2.2 Requirements

ID	2F_05.01.Zefiro-Europe.TRS.120
Description	The applied tractive force shall be compensated with the current weight of the train.
Safety level	SIL 0

The train mass is measured by the secondary suspension, read by the BCU and forwarded to TCMS software and PCU for load compensation of the traction demand.

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Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.70	The car weight as measured by the secondary suspension device shall be forwarded to the TCMS on MVB.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.01.Zefiro-Europe.CONCEPT.633	TCMS shall be notified if the measured weight is invalid.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.01.Zefiro-Europe.CONCEPT.184	The train weight correction factor shall be calculated and sent to each PCU in the train. (to be updated with loadweigh curve for correction factor calculation). (appr. 90 % = service weight and 100 % = normal weight of TSI normal load depending of the bogie weight.)  	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.01.Zefiro-Europe.CONCEPT.618	If the car weight cannot be calculated for a car the train weight correction factor shall be set to 100% for that car, i.e. full load shall be considered.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.01.Zefiro-Europe.CONCEPT.636	If the train weight is considered invalid, an event shall be sent.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_05.01.Zefiro-Europe.CONCEPT.185	The traction demand received from the TCMS shall be considered not load compensated. The compensation shall be calculated internally using the train weight correction factor received from TCMS.	Derived Requirement	4S_05.-.Propulsion		N/A

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3.3 Car ID and Vehicle Type

3.3.1 Function Design

All traction packages in each motor car work independently from each other.

In order to coordinate all these local propulsion packages, the car ID shall be distributed in all motor cars through the train from TCMS.

3.3.1.1 Normal condition

In normal conditions, the car ID is distributed from TCMS to all local PCUs which will check the consistency of the received information.

The car ID will be used by the diagnostic function in the PCU.

3.3.1.2 Failure cases

Description of failure cases

A) The car ID sent to PCU is out of range or mismatch between Car ID and MVB address. Traction will be disabled in each Propulsion module that have invalid car ID.
B) Communication fault between TCMS and PCU. No Car ID will be sent to the PCU.

3.3.1.3 Failure detection

Description of failure detection

A) If the Car ID is not between 1 and 8 or the Car ID can not be mapped to the static MVB address of the PCU.
B) The communication between CCU and the PCU is supervised by the CCU which will trigger an event stored in the diagnostic database.

3.3.1.4 Failure handling

Description of failure handling

A) Fault will be logged in the diagnostic database and checked at maintenance. Unscheduled workshop visit at the earliest after two days if only one propulsion module is affected. Otherwise: Major failure.
B) Fault will be logged in the diagnostic database and checked at maintenance. Unscheduled workshop visit at the earliest after two days if only one propulsion module is affected. Otherwise: Major failure.

3.3.2 Requirements

ID	2F_05.01.Zefiro-Europe.TRS.104
Description	TCMS shall send a car ID to the Propulsion Control (PCU)
Safety level	SIL 0

The traction packages work independently from each other and need to be informed about the car ID from TCMS.

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<table>
<tr><th>Nr</th><th>Description</th><th>Type</th><th>Derived to</th><th>User Interface</th><th>SIL</th></tr>
<tr><td>2F_05.01.Zefiro-Europe.CONCEPT.105</td><td>TCMS shall send a car ID to the PCU according to the table below:
<table><tr><th>Car identity</th><th>MVB addr</th><th>Car description</th></tr><tr><td>1</td><td>40</td><td>DM1</td></tr><tr><td>1</td><td>41</td><td>DM1</td></tr><tr><td>3</td><td>42</td><td>M3</td></tr><tr><td>3</td><td>43</td><td>M3</td></tr><tr><td>6</td><td>63</td><td>M6</td></tr><tr><td>6</td><td>62</td><td>M6</td></tr><tr><td>8</td><td>61</td><td>DM8</td></tr><tr><td>8</td><td>60</td><td>DM8</td></tr></table></td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL 0</td></tr>
<tr><td>2F_05.01.Zefiro-Europe.CONCEPT.619</td><td>If the car ID sent from TCMS is <= 0 or >= 9 or the MVB address of the DCU/L does not correspond to the valid Car ID according to below table then an event shall be set.
<table><tr><th>Car identity</th><th>MVB addr</th><th>Car description</th></tr><tr><td>1</td><td>40</td><td>DM1</td></tr><tr><td>1</td><td>41</td><td>DM1</td></tr><tr><td>3</td><td>42</td><td>M3</td></tr><tr><td>3</td><td>43</td><td>M3</td></tr><tr><td>6</td><td>63</td><td>M6</td></tr><tr><td>6</td><td>62</td><td>M6</td></tr><tr><td>8</td><td>61</td><td>DM8</td></tr><tr><td>8</td><td>60</td><td>DM8</td></tr></table></td><td>Derived Requirement</td><td>4S_05.-.Propulsion</td><td>EVENT</td><td>N/A</td></tr>
</table>

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3.4 MCM Enabling

3.4.1 Function Design

The MCM is enabled if traction shall be applied. The enable conditions are checked by TCMS according to train control parameters (master controller position, train speed).

3.4.1.1 Normal condition

The MCM shall be enabled only when the required conditions are present.
These required enable conditions are checked by TCMS which are then sent via PCU to enable both MCM.

The MCM shall be stopped in some circumstances.
These circumstances are checked by TCMS which will disable both MCM via PCU. For conditions when to enable and disable see below derived requirements.

3.4.1.2 Failure cases

Description of failure cases

A) Communication fault between TCMS and PCU. The MCMs connected to that PCU can not be enabled thus no traction is available from these MCMs.
B) The MCM does not respond to the enable command. Traction will not be available for that MCM.
C) The MCM is enabled when not ordered. The motor will be magnetized even though the train is not intended to request traction. No impact on the operation.

3.4.1.3 Failure detection

Description of failure detection

A) The communication between CCU and the PCU is supervised by the CCU which will trigger an event stored in the diagnostic database.
B) PCU will generate an event that will be displayed on the IDU indicating that the MCM do not start on order.
C) PCU will generate an event that will be displayed on the IDU indicating that the MCM is started without an order.

3.4.1.4 Failure handling

Description of failure handling

A) Fault will be logged in the diagnostic database and checked at maintenance. Unscheduled workshop visit at the earliest after two days.
B) No action defined for the driver. Fault will be checked during maintenance. To be maintained at next scheduled workshop visit.
C) No action defined for the driver. Fault will be checked during maintenance. To be maintained at next scheduled workshop visit.

3.4.2 Requirements

ID	2F_05.01.Zefiro-Europe.TRS.163
Description	The MCM shall be stopped when – traction safe trainline is low OR – Train speed is 0 and Master Controller is not in a traction position – doors are released
Safety level	N/A

ID	2F_05.01.Zefiro-Europe.TRS.89
Description	MCM shall be enabled when the Master Controller is in traction position or train speed is >0.
Safety level	N/A

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Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.115	Each individual MCM in the train (single or multiple) shall be enabled (started) when the Traction Controller is in a traction position OR Train speed is greater than 0 OR Emergency mode is active. The individual MCM shall be disabled (stopped) when – Traction safe trainline is de-energised OR – Train is towed OR – Train speed is 0 and Traction Controller has not been in traction position for at least 2 seconds OR – "Traction drive major mechanical fault" is set OR – MCM is cut out OR – The doors are ordered to release. If both enable and disable conditions are true at the same time the MCMs shall be disabled.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.01.Zefiro-Europe.CONCEPT.204	The PCU shall enable the MCM if: – MVB signal MCM enable received	Derived Requirement	4S_05.-.Propulsion		N/A
2F_05.01.Zefiro-Europe.CONCEPT.622	If the MCM is not started (enabled) when ordered by TCMS then an event shall be generated.	Derived Requirement	4S_05.-.Propulsion	EVENT	N/A
2F_05.01.Zefiro-Europe.CONCEPT.623	If the MCM is started (enabled) when **not** ordered by TCMS then an event shall be generated.	Derived Requirement	4S_05.-.Propulsion	EVENT	N/A

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3.5 LCM Interlacing

3.5.1 Function Design

The phase shift set point is calculated for each line converter in the train according to the number of total active line converters in the train and their relative location. The aim is to minimize the harmonics generated out on the supply line by distributing the phase shift evenly for all active line converters in the train.
The location of each line converter is calculated according to total number of units in the train, unit position in the train, number of active LCMs in the unit and position of the active line converter in respectively unit.
One unit corresponds to one 4-car segment, which implies the train consists of unit 1-and 2. Each unit includes four LCMs 1 to-4 divided into two propulsion systems. In case of one LCM is faulty/cut-out all LCM positions will be recalculated. In multiple operations the train consists of unit 1 to-4 and are divided into four propulsion systems.

3.5.1.1 Normal condition

The train function of interlacing is done by TCMS by notifying the PCU of the position of LCM and the amount of active LCM in the train. A switching angle shall be assigned to every individual LCM.
Though the switching angles of the LCM on the train will be evenly distributed over 360 degrees. This train function of interlacing is achieved by TCMS by notifying the PCU of the position of LCM and the amount of active LCM in the train.

The PCU shall calculate the corresponding phase shift set point for each individual LCM regarding the position of the LCM and the amount of active LCM in the train.

3.5.1.2 Failure cases

Description of failure cases
A) Incorrect position of active LCM in a unit is sent to the PCU. Wrong phase shift set point will be calculated by the PCU. The EMC limit could be exceeded.
B) Incorrect number of active LCMs in unit sent to PCU. Wrong phase shift set point will be calculated by the PCU. The EMC limit could be exceeded.
C) Incorrect unit position in train sent to PCU. Wrong phase shift set point will be calculated by the PCU. The EMC limit could be exceeded.
D) Incorrect total number of units in train sent to the PCU. Wrong phase shift set point will be calculated by the PCU. The EMC limit could be exceeded.
E) Communication fault between TCMS and PCU. The LCM in affected PCU will not be active.

3.5.1.3 Failure detection

Description of failure detection
A) Not detected.
B) Not detected.
C) Not detected.
D) Not detected.
E) The communication between CCU and the PCU is supervised by the CCU which will trigger an event stored in the diagnostic database.

3.5.1.4 Failure handling

Description of failure handling
A) No failure handling.
B) No failure handling.
C) No failure handling.
D) No failure handling.
E) The number of LCMs in unit and the position of LCM in unit will be re-calculated by TCMS and sent to the PCU. The PCU will then re-calculate the phase shift set point for remaining LCMs in the unit. Unscheduled workshop visit at the earliest after two days.

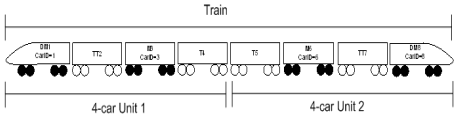
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3.5.2 Requirements

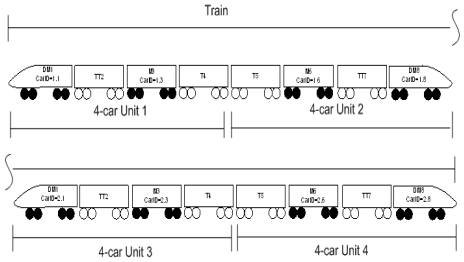
ID	2F_05.01.Zefiro-Europe.TRS.191
Description	Interlacing between line converters shall be provided: the line converters shall switch with shifted pulse pattern between each other. The switching angels shall be distributed evently over the whole train over 360 degrees.
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.239	The following signals shall be calculated and sent to each PCU in the train: – Position of active LCM in unit – Number of active LCMs in unit – Unit position in Train – Total number of units in train The LCM shall be considered active if the status from the LCM is not faulty and the LCM is not cut-out from the IDU and communication is OK between TCMS and the PCU. In single train operation the above signals shall be calculated based on following picture and table:	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 1



Car identity	Total number of Units in Train	Unit position in Train	Number of active LCMs in unit	Position of LCM in unit
1	2	1	4	1
1	2	1	4	3
3	2	1	4	2
3	2	1	4	4
6	2	2	4	4
6	2	2	4	2
8	2	2	4	3
8	2	2	4	1

In multiple train operation the above signals shall be calculated based on following picture and table:



Car identity	Total number of Units in Train	Unit position in Train	Number of active LCMs in unit	Position of LCM in unit
1.1	4	1	4	1
1.1	4	1	4	3
1.3	4	1	4	2
1.3	4	1	4	4
1.6	4	2	4	4
1.6	4	2	4	2
1.8	4	2	4	3
1.8	4	2	4	1
2.1	4	3	4	1
2.1	4	3	4	3
2.3	4	3	4	2
2.3	4	3	4	4
2.6	4	4	4	4
2.6	4	4	4	2
2.8	4	4	4	3
2.8	4	4	4	1

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2F_05.01.Zefiro-Europe.CONCEPT.398	For each LCM that is not in operation the number of LCMs in unit shall be decreased by 1 and all LCM positions with a higher position than the non-available LCM (within the same unit) shall be decreased by 1.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 1
2F_05.01.Zefiro-Europe.CONCEPT.659	For each high voltage unit that is faulty (transformator is isolated) the Number of units shall be decreased by 1 and Unit position for units with a higher position than the non-available unit number shall be decreased by 1.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_05.01.Zefiro-Europe.CONCEPT.252	The PCU shall indicate to TCMS if the LCM is faulty.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_05.01.Zefiro-Europe.CONCEPT.285	PCU shall calculate the switching angle for individual LCM depending on the train wide LCM status report from TCMS.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_05.01.Zefiro-Europe.CONCEPT.667	If only one LCM is in operation on one transformer at normal operation (not startup or short blocking conditions) then the LCM shall be stopped and an event shall be set indicating that the LCM is stopped due to EMC restrictions.	Derived Requirement	4S_05.-.Propulsion		

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3.6 Brake Backup mode

3.6.1 Function Design

Backup braking is a propulsion system function which is used to feed the auxiliary system and regulate the DC link if no line voltage is available. A small electro dynamical braking force is generated by the traction motors to feed energy into the DC link. Charging of the DC-link from the traction motors will also be available when the train is towed by using the pre-charge unit.

3.6.1.1 Normal condition

If the PCU received from TCMS the towing signal or the crossing neutral sections signal, or during interruption of line voltage, the MCM will be working in backup braking mode and will feed the DC link.

3.6.1.2 Failure cases

Description of failure cases

A) Communication fault between TCMS and PCU. Brake back-up mode will not be available.
B) No DC-link power supply due to faulty pre-charge unit.
C) Requested signal not sent to the PCU. The DC-link will not be charged.

3.6.1.3 Failure detection

Description of failure detection

A) The communication between CCU and the PCU is supervised by the CCU which will trigger an event stored in the diagnostic database.
B) PCU supervises the pre-charging unit. No DC link voltage or missing feedback from the pre-charging unit.
C) Probably the towing switch on the driver's desk is faulty. The towing switch is not in the scope of Drive Control SFRS.

3.6.1.4 Failure handling

Description of failure handling

A) Fault will be logged in the diagnostic database and checked at maintenance. Unscheduled workshop visit at the earliest after two days.
B) No backup braking for that Propulsion System. To be maintained at next scheduled workshop visit. An event shall be sent in case of failure.
C) Unknown.

3.6.2 Requirements

ID	2F_05.01.Zefiro-Europe.TRS.151
Description	The MCM shall work in Brake Backup mode to supply auxiliary power and to regulate DC link voltage in the following cases: – Crossing neutral sections – Loss of line voltage – Towing operation
Safety level	N/A

Backup braking is used for the propulsion system to feed and regulate the DC link if no line voltage is available. A small electro dynamical braking force is generated by the traction motors to feed energy into the DC link.

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Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.130	The MCM shall work in Brake Backup mode to supply auxiliary power and to regulate DC link voltage in the following cases: – Crossing neutral sections (status provided by TCMS over MVB) – Interruption of line voltage – Towing operation (status provided by TCMS over MVB)	Derived Requirement	4S_05.-.Propulsion		N/A
2F_05.01.Zefiro-Europe.CONCEPT.210	TCMS shall provide the following status signals to PCU: – towing operation active, this train is "towed" – neutral section passage is active	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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3.7 Boost Mode

3.7.1 Function Design

Boost mode is a propulsion system function used to compensate the reduction of traction power of one part of the train by using extra reserves from the remaining traction units.

3.7.1.1 Normal condition

TCMS will distribute the train status data dependent on the number of active MCM in the train.

The Boost mode shall be enabled when some traction power is lost, the remaining traction units shall be used to compensate this loss as much as possible.

3.7.1.2 Failure cases

Description of failure cases

A) Number of operational MCMs in train is falsely calculated by TCMS. The boost mode will falsely be activated or not activated when requested depending on the number of operational MCMs.
B) Communication between TCMS and PCU is faulty. The other PCUs will compensate for the lost traction in the Propulsion package

3.7.1.3 Failure detection

Description of failure detection

A) None identified.
B) The communication between CCU and the PCU is supervised by the CCU which will trigger an event stored in the diagnostic database.

3.7.1.4 Failure handling

Description of failure handling

A) None.
B) Fault will be logged in the diagnostic database and checked at maintenance. Unscheduled workshop visit at the earliest after two days.

3.7.2 Requirements

ID	2F_05.01.Zefiro-Europe.TRS.74
Description	In case of loss of traction power the remaining traction units shall be used to compensate the reduction as much as possible. If the remaining traction units have sufficient thermal reserves under the current external temperature conditions, the lost traction power shall be compensated.
Safety level	N/A

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_05.01.Zefiro-Europe.CONCEPT.135	In case of any loss of traction power on the train, the remaining traction units shall be used to compensate the reduction as much as possible. If the remaining traction units have sufficient thermal reserves under the current external temperature conditions, the lost traction power shall be compensated. The compensation of traction power shall be done on two levels on command from TCMS.	Derived Requirement	4S_05.-.Propulsion		N/A

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2F_05.01.Zefiro-Europe.CONCEPT.666	Boost mode shall only be set at the following conditions – Ambient temperature is less or equal to 25degC. – Selected system is 25kV AC, 15kV AC or 3 kV DC. – Line voltage is greater or equal to 3.4 kV DC at 3 KV DC system to activate boost mode. Note: This is only a start condition, boost mode shall not be deactivated due to line voltage below 3.4 kV DC.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.01.Zefiro-Europe.CONCEPT.286	"Boost mode 1" shall be set to all PCUs when 1, 2 or 3 MCMs in the train are not available.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.01.Zefiro-Europe.CONCEPT.665	"Boost mode 2" shall be set to all PCUs when 4 MCMs in the train are not available.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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3.8 Slip Slide Control

3.8.1 Function Design

The wheel slide protection and wheel slip protection are used to detect and prevent the wheel-slip during braking and wheel-slide during acceleration by regulating the traction torque depending on the adhesion.

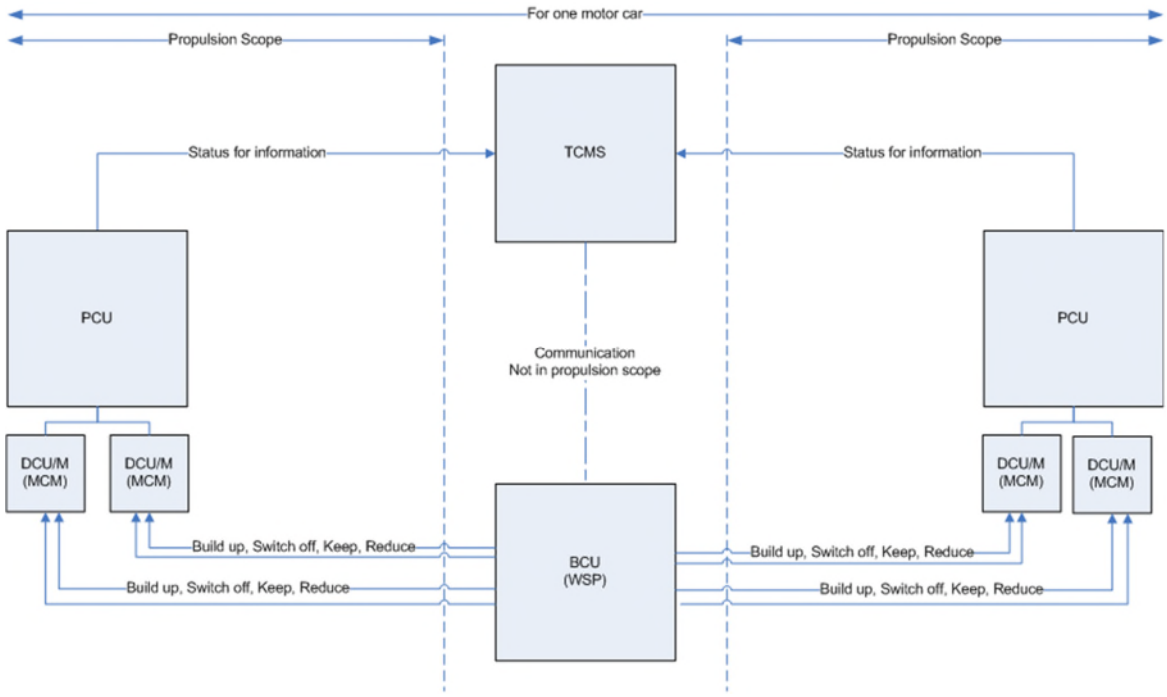
3.8.1.1 Normal condition

This function is based on the communication between Braking System, Propulsion System and TCMS.

Two wired signals are sent by WSP and read by each MCM. The statuses of these signals are also sent to TCMS.

One wired signal "electro dynamic (ED) brake off" shall be sent by WSP and read by each MCM. The statuses of these signals are also sent to TCMS.

TCMS shall send the train speed to PCU. The signal shall be calibrated considering the wheel diameters.



3.8.1.2 Failure cases

Description of failure cases
The WSP control is managed by Braking system. No failure cases will be defined in this document.

3.8.1.3 Failure detection

Description of failure detection
N/A

3.8.1.4 Failure handling

Description of failure handling
N/A

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3.8.2 Requirements

ID	2F_05.01.Zefiro-Europe.TRS.161
Description	The traction axles shall be equipped with an anti-slip system.
Safety level	N/A

ID	2F_05.01.Zefiro-Europe.TRS.201
Description	The traction axles shall be equipped with an anti-slide system, this shall interact with the brake systems wheel side protection (WSP).
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.265	The WSP shall provide two wired signals to PCU (DCU/M) for slide control management with the following codes: – build up – switch off – keep – reduce the status of the signals shall be fed back to TCMS for information.	Derived Requirement	4S_07.-.Braking System		SIL 0
2F_05.01.Zefiro-Europe.CONCEPT.268	PCU shall read two wired signals for slide control management with the following codes: – build up , – switch off – keep – reduce the status of the signals shall be fed back to TCMS for information.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_05.01.Zefiro-Europe.CONCEPT.430	If Emergency brake is active then the PCU shall be notified.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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3.9 Traction Motor Cooling

3.9.1 Function Design

For the cooling of the traction motor there is one traction motor fan for each traction motor.
Each fan is controlled by a variable frequency converter. The set point for the converter is set by PPC via an AX unit.

The converter is supervised via digital interface to a MIO-unit.

3.9.1.1 Normal condition

In normal condition the PCU is controlling the fan via the frequency converter.

3.9.1.2 Failure cases

Description of failure cases
A) Fan fault
B) Frequency converter fault
C) AX unit fault

3.9.1.3 Failure detection

Description of failure detection
A) By temperature supervision of motor.
B) Through status bits from frequency converter.
C) AX unit supervision or Fan is not starting on order

3.9.1.4 Failure handling

Description of failure handling
A) Motor converter will be blocked.
B) Fault will be logged in the diagnostic database and checked at maintenance.
C) Fault will be logged in the diagnostic database and checked at maintenance.

3.9.2 Requirements

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.649	The set point shall be connected from the AX unit (connected to PPC MVB) to the frequency converter for traction motor cooling.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_05.01.Zefiro-Europe.CONCEPT.650	The "start code" to the frequency converter shall be set when ordered from TCMS DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_05.01.Zefiro-Europe.CONCEPT.651	The status bit "MK1" from the frequency converter shall be connected to TCMS DI.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_05.01.Zefiro-Europe.CONCEPT.652	The status bit "MK2" from the frequency converter shall be connected to TCMS DI.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_05.01.Zefiro-Europe.CONCEPT.654	The status bit "MK3" from the frequency converter shall be connected to TCMS DI.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_05.01.Zefiro-Europe.CONCEPT.673	The 3 status bits from frequency converter shall be evaluated to receive the status of the converter.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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2F_05.01.Zefiro-Europe.CONCEPT.655	The "Status Traction motor cooling inverter OK" shall be set to PCU when the frequency converter is not permanently faulty and 400V supply is available for the traction motor cooling fan. Motor Fan OK signal shall be removed in 20 sec, If 400V Missing OR Permanent Fault is active [Mk1=0; Mk2=0; Mk3=0] OR Temporary Fault is active [Mk1=0; Mk2=1; Mk3=1] OR (DBC_ACO_C_ActColMt = 1 (Command Signal From PPC) AND DBC_ACO_S_MK1 = 0 (bit MK1) for more than 10 sec with 400V present and contactor closed (to avoid start-up misalignements))	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.01.Zefiro-Europe.CONCEPT.680	Any time TMF does not start on order, TCMS shall try a new start up resetting "start bit" command by DO. After 3 consecutive missing starts are performed within 60 seconds, an event shall be set and related MCM shall be blocked. Fault count shall be reset anytime 400Vac power supply is reset	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_05.01.Zefiro-Europe.CONCEPT.671	When the status from TCMS "Traction motor cooling inverter ok" is not set the motor converter shall be blocked. Note that the motor converter shall not be blocked due to inverter not ok when motor converter is in backup braking mode, e.g. during towing.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_05.01.Zefiro-Europe.CONCEPT.656	An event shall be set when the converter reports a failure according to the status bits. Permanent Fault shall be reset every time 400 V is not availeble on the consist Permanent fault iis defined if 400 V is provided to the TMF and the bit are all 0 [Mk1=0; Mk2=0; Mk3=0]	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.01.Zefiro-Europe.CONCEPT.672	An event shall be set when the converter does not start on order, i.e. when indication that "motor is working" is not set from the frequency converter when "start code" is set to the frequency converter.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.01.Zefiro-Europe.CONCEPT.670	The command to activate the cooling fan shall be set to TCMS when the reference frequency for the cooling fan is higher than 0.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_05.01.Zefiro-Europe.CONCEPT.657	The start order "start code" to the converter shall be set when requested from PCU and 3AC400V contactor "KTMC" has been set for at least 5s per each Motor car.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_05.01.Zefiro-Europe.CONCEPT.668	The 3AC400V contactor "KTMC" shall be closed when ordered from TCMS DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

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<table>
<tr>
<td>2F_05.01.Zefiro-Europe.CONCEPT.669</td>
<td>oThe DO for closing of 3AC400V contactor "KTMC" of the traction motor cooling shall be set when 400V supply is available and a specific delay, per each Motor car, is elapsed

<table>
<tr><th>Car</th><th>Delay (s)</th></tr>
<tr><td>DM1</td><td>2</td></tr>
<tr><td>M3</td><td>4</td></tr>
<tr><td>M6</td><td>6</td></tr>
<tr><td>DM8</td><td>8</td></tr>
</table>

If the motor fan is in permanent fault for more than 60 sec the related DO for closing of 3AC400V contactor "KTMC" of the traction motor cooling shall be opened. After 60 sec. the DO for closing of 3AC400V contactor "KTMC" of the traction motor cooling shall be closed and an even shall be set “First attempt to restarting motor fan after a permanent fault “. If the motor fan is still in permanent fault after 60 sec. the related DO for closing of 3AC400V contactor "KTMC" of the traction motor cooling shall be opened. After 120 sec. the DO for closing of 3AC400V contactor "KTMC" of the traction motor cooling shall be closed and an even shall be set “Second attempt to restarting motor fan after a permanent fault “. If the motor fan is still in permanent fault after 60 sec. the related DO for closing of 3AC400V contactor "KTMC" of the traction motor cooling shall be opened. After 300 sec. the DO for closing of 3AC400V contactor "KTMC" of the traction motor cooling shall be closed and an even shall be set “Third attempt to restarting motor fan after a permanent fault “. If after 60 sec. the motor fan is still in permanent fault the motor fan shall be blocked (no restart are allowed until permanent fault is set to zero) and an event shall be set “To many permanent fault of the motor fan”.
If during this sequence the 400 V is not available, the sequence shall be stopped and the normal sequence shall be activated.</td>
<td>Derived Requirement</td>
<td>4S_09.03.05.-.TCMS Software</td>
<td></td>
<td>N/A</td>
</tr>
</table>

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3.10 Commissioning and Test

3.10.1 Activation of Transient Recorder

3.10.1.1 Function Design

In order to achieve some tests and commissioning, the transient recorders within the propulsion equipment in the train shall be activated centrally from one CCU.

3.10.1.1.1 Normal condition

All transient recorders from each PCU shall be activated by receiving a trigger signal from TCMS.

3.10.1.1.2 Failure cases

Description of failure cases
The configuration and activation of the transient recorder is usually done during advanced fault finding. Identifying failures of function is not applicable because a failure will not have any impact on the operation of the train nor affect normal maintenance.

3.10.1.1.3 Failure detection

Description of failure detection
N/A

3.10.1.1.4 Failure handling

Description of failure handling
N/A

3.10.1.2 Requirements

ID	2F_05.01.Zefiro-Europe.TRS.97
Description	For commissioning and tests it shall be possible to activate all transient recorders within the Propulsion equipment in the train by setting a parameter in the TCMS
Safety level	N/A

In order to achieve some tests and commissioning, the transient recorders within the propulsion equipment in the train shall be activated

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.140	Activation of all transient recorders within the Propulsion equipment shall be performed by receiving a trigger signal on MVB or IP (via DCUTerm session directly connected).	Derived Requirement	4S_05.-.Propulsion		N/A
2F_05.01.Zefiro-Europe.CONCEPT.211	A simple parameter (not stored in NVRAM) shall be defined to set all propulsion related transient recorders to active.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.10.2 Cut Out of Converter

3.10.2.1 Function Design

During commissioning and tests, it shall be possible to cut out (disable) all converter modules by pressing a soft button on IDU. This function shall only be accessible for maintenance personal.

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3.10.2.1.1 Normal condition

From the IDU of the maintenance personal, it shall be presented a menu for cut out of every converter module per motor car.

From a signal on MVB, it shall be possible to cut out every converter module individually.

3.10.2.1.2 Failure cases

Description of failure cases

A) Communication fault between TCMS and PCU. The LCM, ACM, FCM and MCMs controlled by the PCU will not be possible to be cut-out from the IDU.
B) Converter is not cut-out when order by TCMS.
C) Converter is cut-out or static cut-out when no order is present from TCMS.

3.10.2.1.3 Failure detection

Description of failure detection

A) The communication between CCU and the PCU is supervised by the CCU which will trigger an event stored in the diagnostic database.
B) An event will be generated and displayed on the IDU indicating that the converter does not cut-out on order.
C) An event will be generated and displayed on the IDU indicating that the converter is cut-out even though no cut-out order is present.

3.10.2.1.4 Failure handling

Description of failure handling

A) Fault will be logged in the diagnostic database and checked at maintenance.
B) Fault will be checked at maintenance.
C) Fault will be checked at maintenance.

3.10.2.2 Requirements

ID	2F_05.01.Zefiro-Europe.TRS.91
Description	For commissioning and tests it shall be possible to cut-out each LCM, ACM and MCM (via IDU)
Safety level	N/A

All converter modules can be cut out (disabled) by pressing a soft button on IDU. This function shall only be accessible for maintenance personal.

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.145	The IDU shall provide a menu for cut out (disabling) of the following converter modules individually per motor car: – 4 MCM – 2 LCM – 2 FCM – 1 ACM The cut-out status of each converter shall remain when the train is deactivated.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_05.01.Zefiro-Europe.CONCEPT.212	It shall be possible to cut out all of the converter modules individually upon a request indicated by MVB signals.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_05.01.Zefiro-Europe.CONCEPT.624	If the converter does not respond to the cut-out order from TCMS then an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_05.01.Zefiro-Europe.CONCEPT.625	If the converter is cut-out but no cut-out order is present from TCMS then an event shall be generated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

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3.11 Maintenance Tests

3.11.1 Automatic Test

3.11.1.1 Function Design

The automatic propulsion system tests shall be possible by triggering on IDU by the maintenance personal.

3.11.1.1.1 Normal condition

The automatic propulsion system test shall be triggered by a soft button on IDU by a maintainer.

The automatic test can be performed under the precondition that the train is at standstill. Then the following sequence shall be performed:

1. Apply full service brake in train
2. Set Traction Block
3. Send Automatic test command to all PCU units in the train
4. Collect test status from all PCU after the test
5. Reset Traction Block (note: only test condition)

Automatic test is activated and forwarded by MVB signal from TCMS.

PCU shall perform an automatic test.

PCU shall report the test status.
The results of the automatic test shall be reported via TDS events.

3.11.1.1.2 Failure cases

Description of failure cases

A) Communication fault between TCMS and PCU. The automatic test of the Propulsion module can not be tested.
B) The automatic test is not started when requested by TCMS even though conditions for test were fulfilled.
C) The automatic test is started when not requested by TCMS.

3.11.1.1.3 Failure detection

Description of failure detection

A) The communication between CCU and the PCU is supervised by the CCU which will trigger an event stored in the diagnostic database.
B) The PCU will generate an event (part of the standard product) if the automatic test can not be started when requested.
C) TCMS will supervise the "test in progress" signals sent from the PCU. If the automatic test is started when not requested protective shutdown will be requested to the PCU.

3.11.1.1.4 Failure handling

Description of failure handling

A) Fault will be logged in the diagnostic database and checked during maintenance.
B) Fault will be logged in the diagnostic database and checked during maintenance.
C) No action from maintenance staff. It will be notified that the Propulsion module has been disabled.

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3.11.1.2 Requirements

ID	2F_05.01.Zefiro-Europe.TRS.99
Description	It shall only be possible to start the Automatic test when – Train speed is zero AND – Maintainer logged in on IDU
Safety level	N/A

ID	2F_05.01.Zefiro-Europe.TRS.167
Description	When the Automatic test is activated the following sequence shall manage the test: 1. Apply full service brake in train (equivalent to Brake Position 7 of Master Controller) 2. Set Traction Block 3. Send Automatic test command to all propulsion control's in the train 4. Collect test status from all propulsion control's 5. Reset Traction Block (note: only test condition)
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_05.01.Zefiro-Europe.CONCEPT.150	When the Automatic test is activated as indicated by MVB signal from TCMS, the PCU shall perform an automatic test and report the test status after finalisation of the tests to TCMS using process data signals. The automatic test shall not be started if the speed is greater than 0. The automatic test shall be aborted if the speed is greater than 0.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_05.01.Zefiro-Europe.CONCEPT.289	The results of the automatic test shall be reported via TDS events. Events related to the automatic test are part of the standard product.	Derived Requirement	4S_05.-.Propulsion	EVENT	N/A
2F_05.01.Zefiro-Europe.CONCEPT.213	The automatic propulsion system test shall be triggered by a soft button on IDU by a maintainer under the preconditions that the train is at standstill, indipendently from the status of HV. Then the following sequence shall be performed: 1. Apply full service brake in train 2. Set Traction Block 3. Send Automatic test command to all PCU units in the train 4. Wait for all PCUs to finish the test. 5. If all PCUs indicates test ok then test passed shall be shown on the IDU. If one or more PCUs indicates test failed then test failed shall be shown on the IDU (one event in case the test has been perfomed in in Low Voltage, one event in case the test has been performed in HV) 6. Reset Traction Block (note: only test condition) 7. Reset full service brake	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_05.01.Zefiro-Europe.CONCEPT.214	The IDU shall provide a soft button for automatic traction system test. This button shall only be accessible in maintenance mode, indipendently from the status of HV, i.e. by workshop personnel logged in as maintainer.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_05.01.Zefiro-Europe.CONCEPT.628	If the PCU indicates that the automatic test is in progress but the test has not been requested from the IDU then protective shutdown shall be requested to the PCU. In this situation an event shall be generated to be displayed on the IDU for the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

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2F_05.01.Zefiro-Europe.CONCEPT.678	The IDU shall provide a soft button for line trip test. This button shall only be accessible in maintenance mode, i.e. by workshop personnel logged in as maintainer.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_05.01.Zefiro-Europe.CONCEPT.679	The results of the line trip test shall be reported via TDS events.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

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3.12 Condition Based Maintenance

3.12.1 CBM test start

The CBM test shall be command by TCMS to propulsion if the following conditions are valid:

1) test is command by the maintainer via IDU
2) the train has to be not in parking mode

If more than 168 hours are passed from the last test, an event shall be set from TCMS to the maintainer

3.12.2 Test

To activate the test to PPC, TCMS has also to send the variables below:

CCU-O - > PCU	Signal	Cycle time	Description	Scale
CT1CX403	CI_TCMS_XU_Bat	512 ms	Battery voltage	10=1V
CT1CX4C1:8	CI_TCMS_VU_Bat	512 ms	Valid bit: Battery voltage	L
CT1CX3C4:3	CI_TCMS_C_CbmTst	64 ms	Command: CBM test	L

the variable CI_TCMS_C_CbmTst has to be at one during the complete test

To set the variable CI_TCMS_C_CbmTst, TCMS has also to supervise if the following condition are true:

- Train at standstill
- 3-phase bus active
- DC-link are charged, all CMM or LCM are in operation

During the test the conditions

- 3-phase bus active
- DC-link are charged, all CMM or LCM are in operation

Must not be take in consideration to abort the test

Once the test is finished , PPC will set to zero the variable below

PCU - > CCU-O	Signal	Cycle time	Description	Scale
CX1CT4C2:14	CBM_S_CbmTst	512 ms	Status: CBM test ongoing	L

If the test is not passed, PPC will set an event PPC CBM_E_CbmTstFl, this event has to set to the maintainer.
Once the test is completed and not fault event is set, the test is completed, TCMS shall set an event for the maintainer, a massage under the virtual push button has to appear with test completed.
TCMS has to abord the test if after 10 s CI_TCMS_C_CbmTst is 1, CBM_S_CbmTst is still 0. TCMS has to send an event to the maintainer that the CBM test has not started, a massaga under the virtual push button has to appear with test not started.
TCMS has to abord the test if after 130 s CI_TCMS_C_CbmTst is 1, CBM_S_CbmTst is still 1. TCMS has to send an event to the maintainer that the CBM test takes to long time, a massage under the virtual push button has to appear with test to long time.
During the complete test phase a massage has to be set on the IDU test page, that traction is not allowed due to the CBM test

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4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system (propulsion)

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_05.01.Zefiro-Europe.CONCEPT.431	WSP control signal 1 for axle 1	L	2F_05.01.Zefiro-Europe.CONCEPT.268
2F_05.01.Zefiro-Europe.CONCEPT.432	WSP control signal 2 for axle 1	L	2F_05.01.Zefiro-Europe.CONCEPT.268
2F_05.01.Zefiro-Europe.CONCEPT.433	WSP control signal 1 for axle 2	L	2F_05.01.Zefiro-Europe.CONCEPT.268
2F_05.01.Zefiro-Europe.CONCEPT.434	WSP control signal 2 for axle 2	L	2F_05.01.Zefiro-Europe.CONCEPT.268
2F_05.01.Zefiro-Europe.CONCEPT.435	WSP control signal 1 for axle 3	L	2F_05.01.Zefiro-Europe.CONCEPT.268
2F_05.01.Zefiro-Europe.CONCEPT.436	WSP control signal 2 for axle 3	L	2F_05.01.Zefiro-Europe.CONCEPT.268
2F_05.01.Zefiro-Europe.CONCEPT.437	WSP control signal 1 for axle 4	L	2F_05.01.Zefiro-Europe.CONCEPT.268
2F_05.01.Zefiro-Europe.CONCEPT.438	WSP control signal 2 for axle 4	L	2F_05.01.Zefiro-Europe.CONCEPT.268

Outputs:

4.1.2 Supplier system (brake)

Inputs:

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_05.01.Zefiro-Europe.CONCEPT.439	WSP control signal 1 for axle 1	L	2F_05.01.Zefiro-Europe.CONCEPT.265
2F_05.01.Zefiro-Europe.CONCEPT.440	WSP control signal 2 for axle 1	L	2F_05.01.Zefiro-Europe.CONCEPT.265
2F_05.01.Zefiro-Europe.CONCEPT.441	WSP control signal 1 for axle 2	L	2F_05.01.Zefiro-Europe.CONCEPT.265
2F_05.01.Zefiro-Europe.CONCEPT.442	WSP control signal 2 for axle 2	L	2F_05.01.Zefiro-Europe.CONCEPT.265
2F_05.01.Zefiro-Europe.CONCEPT.443	WSP control signal 1 for axle 3	L	2F_05.01.Zefiro-Europe.CONCEPT.265
2F_05.01.Zefiro-Europe.CONCEPT.444	WSP control signal 2 for axle 3	L	2F_05.01.Zefiro-Europe.CONCEPT.265
2F_05.01.Zefiro-Europe.CONCEPT.445	WSP control signal 1 for axle 4	L	2F_05.01.Zefiro-Europe.CONCEPT.265

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2F_05.01.Zefiro-Europe.CONCEPT.446	WSP control signal 2 for axle 4	L	2F_05.01.Zefiro-Europe.CONCEPT.265

4.1.3 Supplier system (drivers desk)

Inputs:

Outputs:

4.1.4 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_05.01.Zefiro-Europe.CONCEPT.661	Status bit MK1		2F_05.01.Zefiro-Europe.CONCEPT.655 2F_05.01.Zefiro-Europe.CONCEPT.673
2F_05.01.Zefiro-Europe.CONCEPT.662	Status bit MK2		2F_05.01.Zefiro-Europe.CONCEPT.656 2F_05.01.Zefiro-Europe.CONCEPT.673
2F_05.01.Zefiro-Europe.CONCEPT.663	Status bit MK3		2F_05.01.Zefiro-Europe.CONCEPT.651 2F_05.01.Zefiro-Europe.CONCEPT.673

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_05.01.Zefiro-Europe.CONCEPT.664	Start code (traction motor cooling inverter)		2F_05.01.Zefiro-Europe.CONCEPT.657

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4.2 Bus Interface

TCMS SW to Propulsion supplier:

Nr	Functional name	Type	Linked to requirement
2F_05.01.Zefiro-Europe.CONCEPT.403	Car identity	A	2F_05.01.Zefiro-Europe.CONCEPT.105
2F_05.01.Zefiro-Europe.CONCEPT.404	LCM configuration data: – Position of active LCM in unit – Number of active LCMs in unit – Unit position in train – Total number of units in train	A	2F_05.01.Zefiro-Europe.CONCEPT.239
2F_05.01.Zefiro-Europe.CONCEPT.405	Neutral section active	L	2F_05.01.Zefiro-Europe.CONCEPT.210
2F_05.01.Zefiro-Europe.CONCEPT.407	Enable start of motor converter	L	2F_05.01.Zefiro-Europe.CONCEPT.204
2F_05.01.Zefiro-Europe.CONCEPT.408	Enable towing mode	L	2F_05.01.Zefiro-Europe.CONCEPT.210
2F_05.01.Zefiro-Europe.CONCEPT.409	Cut-out LCM	L	2F_05.01.Zefiro-Europe.CONCEPT.145
2F_05.01.Zefiro-Europe.CONCEPT.448	Cut-out ACM	L	2F_05.01.Zefiro-Europe.CONCEPT.145
2F_05.01.Zefiro-Europe.CONCEPT.449	Cut-out MCM1	L	2F_05.01.Zefiro-Europe.CONCEPT.145
2F_05.01.Zefiro-Europe.CONCEPT.450	Cut-out MCM2	L	2F_05.01.Zefiro-Europe.CONCEPT.145
2F_05.01.Zefiro-Europe.CONCEPT.451	Cut-out FCM	L	2F_05.01.Zefiro-Europe.CONCEPT.145
2F_05.01.Zefiro-Europe.CONCEPT.421	Train weight correction factor	A	2F_05.01.Zefiro-Europe.CONCEPT.184
2F_05.01.Zefiro-Europe.CONCEPT.422	Emergency brake is active	L	2F_05.01.Zefiro-Europe.CONCEPT.430
2F_05.01.Zefiro-Europe.CONCEPT.630	Start Automatic test	L	2F_05.01.Zefiro-Europe.CONCEPT.150
2F_05.01.Zefiro-Europe.CONCEPT.631	Driving direction backward	L	2F_05.01.Zefiro-Europe.CONCEPT.207
2F_05.01.Zefiro-Europe.CONCEPT.632	Driving direction forward	L	2F_05.01.Zefiro-Europe.CONCEPT.207
2F_05.01.Zefiro-Europe.CONCEPT.634	Transient recorder activation trigg	L	2F_05.01.Zefiro-Europe.CONCEPT.140
2F_05.01.Zefiro-Europe.CONCEPT.677	Traction motor cooling inverter ok		2F_05.01.Zefiro-Europe.CONCEPT.655 2F_05.01.Zefiro-Europe.CONCEPT.671

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Propulsion supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_05.01.Zefiro-Europe.CONCEPT.33	Converter status: LCM is in operation ACM is in operation FCM is in operation MCM1 is in operation MCM2 is in operation LCM is cut-out ACM is cut-out FCM is cut-out MCM1 is cut-out MCM2 is cut-out LCM is faulty ACM is faulty FCM is faulty MCM1 is faulty MCM2 is faulty	L	2F_05.01.Zefiro-Europe.CONCEPT.252
2F_05.01.Zefiro-Europe.CONCEPT.626	Mechanical fault MCM1	L	2F_05.01.Zefiro-Europe.CONCEPT.615
2F_05.01.Zefiro-Europe.CONCEPT.627	Mechanical fault MCM2	L	2F_05.01.Zefiro-Europe.CONCEPT.615
2F_05.01.Zefiro-Europe.CONCEPT.629	Automatic test in progress	L	2F_05.01.Zefiro-Europe.CONCEPT.628
2F_05.01.Zefiro-Europe.CONCEPT.660	Transformer is isolated		2F_05.01.Zefiro-Europe.CONCEPT.659
2F_05.01.Zefiro-Europe.CONCEPT.676	Activate cooling for Traction motor		2F_05.01.Zefiro-Europe.CONCEPT.657 2F_05.01.Zefiro-Europe.CONCEPT.670

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5 References

TRD Propulsion system: 3EGH489007-6600
ICD Interface Control Document: 3EST000225-5108

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6 Revision Log

6.1 Revision log 3EST000225-7909_A

Revision:	**_A**		
Date:	**2019-12-03**		
Editor:	**O. Di Domenico**		
Change ID DOORS	**Refers to requirement ID**	**Title**	**Change content**
2F_05.01.ISSUES.-.Zefiro-Europe.1	[2F_05.01.Zefiro-Europe.CONCEPT.135]: [2F_05.01.Zefiro-Europe.CONCEPT.286]:	Renaming number of Non Faulty MCMs to PCU	For TCMS requirement change 'non-faulty MCMs' to 'available MCMs' to PCU. For Propulsion req. change to: "In case of any loss of traction power on the train, the remaining traction units shall be used to compensate the reduction as much as possible. If the remaining traction units have sufficient thermal reserves under the current external temperature conditions, the lost traction power shall be compensated. The calculation of compensated traction power shall be based on number of available MCMs in the train sent from TCMS."
2F_05.01.ISSUES.-.Zefiro-Europe.2	[2F_05.01.Zefiro-Europe.CONCEPT.213]:	Better description of automatic propulsions system test	Not clear what TCMS shall do in step 4 and service break is not restored when the test is finished. Suggestion to rewrite requirement. "The automatic propulsion system test shall be triggered by a soft button on IDU by a maintainer under the precondition that the train is at standstill. Then the following sequence shall be performed: 1. Apply full service brake in train 2. Set Traction Block 3. Send Automatic test command to all PCU units in the train 4. Wait for all PCUs to finish the test. 5. If all PCUs indicates test ok then test passed shall be shown on the IDU. If one or more PCUs indicates test failed then test failed shall be shown on the IDU. 6. Reset Traction Block (note: only test condition) 7. Reset full service break
2F_05.01.ISSUES.-.Zefiro-Europe.3	[2F_05.01.Zefiro-Europe.CONCEPT.239]: [2F_05.01.Zefiro-Europe.CONCEPT.398]: [2F_05.01.Zefiro-Europe.CONCEPT.659]: [2F_05.01.Zefiro-Europe.CONCEPT.660]:	Recalculation of LCM position if a LCM(s) is faulty	A new derived requirement is needed: If x number of LCM is not available (faulty or cut out) all LCM positions with a higher position than the non-available LCM(s) number (within the same unit) shall be decreased by x. 2012-04-25/AS Proposal: If x number of LCM is not available (faulty or cut out) all LCM positions with a higher position than the non-available LCM(s) number (within the same unit) shall be decreased by x. If a unit is faulty (for example transformator is faulty) the Number of units shall be decreased by 1 and Unit position for units with a higher position than the non-available unit number shall be decreased by 1.

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Revision:	**_A**		
Date:	**2019-12-03**		
Editor:	**O. Di Domenico**		
Change ID DOORS	**Refers to requirement ID**	**Title**	**Change content**
2F_05.01.ISSUES.-.Zefiro-Europe.6	[2F_05.01.Zefiro-Europe.CONCEPT.185]: [2F_05.01.Zefiro-Europe.CONCEPT.207]: [2F_05.01.Zefiro-Europe.CONCEPT.252]: [2F_05.01.Zefiro-Europe.CONCEPT.268]: [2F_05.01.Zefiro-Europe.CONCEPT.285]: [2F_05.01.Zefiro-Europe.CONCEPT.619]: [2F_05.01.Zefiro-Europe.CONCEPT.622]: [2F_05.01.Zefiro-Europe.CONCEPT.623]: [2F_05.01.Zefiro-Europe.CONCEPT.624]: [2F_05.01.Zefiro-Europe.CONCEPT.625]:	Removal of SIL for all Propulsion requirements.	It has been agreed between MLM safety and PPC safety that the safety requirements will be defined by seperate document similar to what was done for the Talent2 project. The SIL shall be set to N/A for all Propulsion requirements.
2F_05.01.ISSUES.-.Zefiro-Europe.4	[2F_05.01.Zefiro-Europe.CONCEPT.115]:	Update of MCM enabling from TCMS	Update requirement to: Each individual MCM in the train (single or multiple) shall be enabled (started) when the Traction Controller is in a traction position OR Train speed is greater than 0 OR Emergency mode is active. The individual MCM shall be disabled (stopped) when – all MCMs in the train indicates that traction safe loop is de-energised OR – Train is towed OR – Train speed is 0 and no traction demand greater than 0% has been given for at least 2 seconds. OR "Traction drive major mechanical fault" is set OR MCM is cut out. If both enable and disable conditions are true at the same time the MCMs shall be disabled. 2012-02-15/AM: The above proposed change is not identical to what is done for the corresponding TCMS Sw TRS requirement. This must be aligned before proceeding any further. 2012-03-12/AS General is no need for identical req. Why then have different levels of req? Clarification is always possible on lower level. SFRS and TRS is now updated.

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Revision:	**_A**
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Editor:	**O. Di Domenico**

Change ID DOORS	Refers to requirement ID	Title	Change content
2F_05.01.ISSUES.-.Zefiro-Europe.7	[2F_05.01.Zefiro-Europe.CONCEPT.637]: [2F_05.01.Zefiro-Europe.CONCEPT.644]: [2F_05.01.Zefiro-Europe.CONCEPT.645]: [2F_05.01.Zefiro-Europe.CONCEPT.646]: [2F_05.01.Zefiro-Europe.CONCEPT.647]: [2F_05.01.Zefiro-Europe.CONCEPT.648]: [2F_05.01.Zefiro-Europe.CONCEPT.649]: [2F_05.01.Zefiro-Europe.CONCEPT.650]: [2F_05.01.Zefiro-Europe.CONCEPT.651]: [2F_05.01.Zefiro-Europe.CONCEPT.652]: [2F_05.01.Zefiro-Europe.CONCEPT.654]: [2F_05.01.Zefiro-Europe.CONCEPT.655]: [2F_05.01.Zefiro-Europe.CONCEPT.656]: [2F_05.01.Zefiro-Europe.CONCEPT.657]: [2F_05.01.Zefiro-Europe.CONCEPT.661]: [2F_05.01.Zefiro-Europe.CONCEPT.662]: [2F_05.01.Zefiro-Europe.CONCEPT.663]: [2F_05.01.Zefiro-Europe.CONCEPT.664]:	Traction motor cooling	Traction motor cooling. These DI and DO to/from TCMS needs to be added and what shall be done with these I/O DI: in order code (ready for operation code, when 400V) DI: operating notification DI: failure message DO: Start code (binary input to VFC) This AO is within PPC scope (propulsion) AO: Fan speed 0-20mA (4-20mA) Proposal: Create new chapter for traction motor cooling. Description: For the cooling of the traction motor there is one traction motor fan for each traction motor. Each fan is controlled by a variable frequency converter. The set point for the converter is set by PPC via an AX unit The converter is supervised via digtal interface to a MIO-unit. Concept: Derive to CTC The set point shall be connected from the AX unit (connected to PPC MVB) to the frequency converter for traction motor cooling. The "start code" to the frequency converter shall be set when ordered from TCMS DO. The "operating notification" from the frequency converter shall be connected to TCMS DI. The "in order code" from the frequency converter shall be connected to TCMS DI. The "failure message" from the frequency converter shall be connected to TCMS DI. Concept: Derive to TCMS The "Status Traction motor cooling inverter OK" shall be set to PCU when the converter is ready for operation (in order code) is set. An event shall be set when the converter is faulty, i.e. the signal "failure message" is set. The start order "start code" to the converter shall be set when requested from PCU.
2F_05.01.ISSUES.-.Zefiro-Europe.9	[2F_05.01.Zefiro-Europe.CONCEPT.105]: [2F_05.01.Zefiro-Europe.CONCEPT.619]:	Car ID	Car ID updated in PPC ICD. Table for car ID needs to be updated acc to PPC ICD.

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Revision:	**_A**		
Date:	**2019-12-03**		
Editor:	**O. Di Domenico**		
Change ID DOORS	**Refers to requirement ID**	**Title**	**Change content**
2F_05.01.ISSUES.-.Zefiro-Europe.10	[2F_05.01.Zefiro-Europe.CONCEPT.184]:	load weight correction	Update with actual values when known.
2F_05.01.ISSUES.-.Zefiro-Europe.11	[2F_05.01.Zefiro-Europe.CONCEPT.647]: [2F_05.01.Zefiro-Europe.CONCEPT.648]: [2F_05.01.Zefiro-Europe.CONCEPT.668]: [2F_05.01.Zefiro-Europe.CONCEPT.669]:	Traction motor cooling failure cases	Traction motor cooling failure cases etc to be added when info is received from Hans Bendien Second question: Is 400V contactor KTMC needed (see visio)? If yes: Add requirement for TCMS control and DO. 2012-08-16/AS: req for 400V contactor is added. 2012-11-29/AS Updated with info from Heinrich Hartmann
2F_05.01.ISSUES.-.Zefiro-Europe.12	[2F_05.01.Zefiro-Europe.CONCEPT.135]: [2F_05.01.Zefiro-Europe.CONCEPT.286]: [2F_05.01.Zefiro-Europe.CONCEPT.558]: [2F_05.01.Zefiro-Europe.CONCEPT.560]: [2F_05.01.Zefiro-Europe.CONCEPT.665]: [2F_05.01.Zefiro-Europe.CONCEPT.666]:	Boost mode	Boost mode needs to be updated (see 4s TRS) Discuss with Frank Unger-Weber 2012-05-31/AS Question clarified with Frank Unger-Weber 2012-06-07/AS Final decisions after meeting with Frank Unger-Weber, Anders Skytt and PPC (Hans R, Patrik E). – Will be considered for AC 25 kV, AC 15 kV and DC 3 kV – Will be implemented for 12.5 % and 25 % loss of traction. – Temperature will be 25 °C or lower – Loose of 1, 2 and 3 MCM's: boost mode characteristic for 87.5 % traction – Loose of 4 MCM's: usage of 75% boost mode characteristic. – Loose of more than 4 MCM's: use normal characteristics (linear reduction ref. to available MCM's) – TCMS start condition at 3 kV DC GE 3.4kV Proposal: "Boost mode 1" shall be set to all PCUs when 1, 2 or 3 MCMs in the train are not available. "Boost mode 2" shall be set to all PCUs when 4 MCMs in the train are not available. Boost mode shall only be set at the following conditions – Ambient temperature LE 25degC. – Selected system is 25kV AC, 15kV AC or 3 kV DC. – Line voltage is GE 3.4 kV DC at 3 KV DC system to ativate boost mode. Note: This is only a start condition, boost mode shall not be deactivated due to line voltage below 3.4 kV DC.

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Revision:	**_A**		
Date:	**2019-12-03**		
Editor:	**O. Di Domenico**		
Change ID DOORS	**Refers to requirement ID**	**Title**	**Change content**
2F_05.01.ISSUES.-.Zefiro-Europe.13	[2F_05.01.Zefiro-Europe.CONCEPT.602]: [2F_05.01.Zefiro-Europe.CONCEPT.624]: [2F_05.01.Zefiro-Europe.CONCEPT.625]:	Event for Cut out of converter	From 4s_TRS_Propulsion CBC protocol (ID 940 and 941). 602: Update description 2012-05-25/FCCB decision to keep req but to derive to TCMS instead of Propulsion.
2F_05.01.ISSUES.-.Zefiro-Europe.14	[2F_05.01.Zefiro-Europe.CONCEPT.619]:	Table MVB address	From 4s_TRS_Propulsion CBC protocol (ID 914). Update table according to concept 105. Proposal **Car identity MVB addr Car description** 1 40 DM1 1 41 DM1 3 42 M3 3 43 M3 6 63 M6 6 62 M6 8 61 DM8 8 60 DM8
2F_05.01.ISSUES.-.Zefiro-Europe.15	[2F_05.01.Zefiro-Europe.CONCEPT.618]:	Load corr factor at sensor fault	When one load sensor is faulty the other sensor should be used. If both sensors in a car are faulty the car should be considered full weight (normal load value) to make sure that 100% traction is received for that car in the calculation of load correction factor. Proposal If the car weight can not be calculated for a car the train weight correction factor shall be set to 100% for that car.
2F_05.01.ISSUES.-.Zefiro-Europe.16	[2F_05.01.Zefiro-Europe.CONCEPT.234]: [2F_05.01.Zefiro-Europe.CONCEPT.667]:	EMC Only one LCM	Due to EMC restrictions it is not allowed to run with only one LCM active on a trafo. Proposal to add after 285: If only one LCM is in operation at normal operation (not startup or short blocking conditions) then the LCM shall be stopped and an event shall be set indicating that the LCM is stopped due to EMC restrictions.

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2F_05.01.ISSUES.-.Zefiro-Europe.18	[2F_05.01.Zefiro-Europe.CONCEPT.670]: [2F_05.01.Zefiro-Europe.CONCEPT.671]: [2F_05.01.Zefiro-Europe.CONCEPT.672]:	Traction motor cooling PCU requirements	Traction motor cooling: Requirement for PCU needs to be added. New concept, derive to propulsion: The command to activate the cooling fan shall be set to TCMS when the reference frequency for the cooling fan is higher than 0. New concept, derive to propulsion: When the status from TCMS "Traction motor cooling inverter ok" is reset the motor converter shall be blocked. Note that the motor converter shall not be blocked due to inverter not ok when motor converter is in backup braking mode, e.g. during towing. New concept derived to TCMS An event shall be set when the converter does not start on order, i.e. when the signal "operating notification" is not set from the frequency converter when "start code" is set to the frequency converter.
2F_05.01.ISSUES.-.Zefiro-Europe.19	[2F_05.01.Zefiro-Europe.CONCEPT.651]: [2F_05.01.Zefiro-Europe.CONCEPT.652]: [2F_05.01.Zefiro-Europe.CONCEPT.654]: [2F_05.01.Zefiro-Europe.CONCEPT.655]: [2F_05.01.Zefiro-Europe.CONCEPT.656]: [2F_05.01.Zefiro-Europe.CONCEPT.661]: [2F_05.01.Zefiro-Europe.CONCEPT.662]: [2F_05.01.Zefiro-Europe.CONCEPT.663]: [2F_05.01.Zefiro-Europe.CONCEPT.672]:	Traction motor cooling status bits	Status signals from cooling converter are changed. Instead of 1 bit for each status TCMS needs to combine the 3 bits to read a status. The requirements needs to be updated (or added) to reflect this.
2F_05.01.ISSUES.-.Zefiro-Europe.20	[2F_05.01.Zefiro-Europe.CONCEPT.678]: [2F_05.01.Zefiro-Europe.CONCEPT.679]:	Line trip test	Line trip test (moved from Issue327 in line voltage) Add two requirements in chapter 3.11 and derive to TCMS – The IDU shall provide a soft button for line trip test. This button shall only be accessible in maintenance mode, i.e. by workshop personnel logged in as maintainer. – The results of the line trip test shall be reported via TDS events. Check if description (information) text needs to be updated.

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2F_05.01.ISSUES.-.Zefiro-Europe.21	[2F_05.01.Zefiro-Europe.CONCEPT.655]:	Traction Motor Fan: faulty status to PPC	Motor Fan OK signal shall be removed in 20 sec, If 400V Missing OR Permanent Fault is active [Mk1=0; Mk2=0; Mk3=0] OR Temporary Fault is active [Mk1=0; Mk2=1; Mk3=1] OR (DBC_ACO_C_ActColMt = 1 (Command Signal From PPC) AND DBC_ACO_S_MK1 = 0 (bit MK1) for more than 10 sec with 400V present and contactor closed (to avoid start-up misalignments))
2F_05.01.ISSUES.-.Zefiro-Europe.22	[2F_05.01.Zefiro-Europe.CONCEPT.619]:	Remove traction block if car ID is wrong	Remove traction block if car ID is wrong – Today PPC does not implement the requ – PPC states that car ID (not a safe telegram) is used only for diagnostic
2F_05.01.ISSUES.-.Zefiro-Europe.23	[2F_05.01.Zefiro-Europe.CONCEPT.680]:	Traction motor fan start attempts.	Any time TMF does not start on order, TCMS shall try a new start up resetting "start bit" command by DO. After 3 consecutive missing starts are performed within 60 seconds, an event shall be set and related MCM shall be blocked. Fault count shall be reset anytime 400Vac power supply is reset
2F_05.01.ISSUES.-.Zefiro-Europe.24	[2F_05.01.Zefiro-Europe.CONCEPT.145]:	Enable cut out of traction components to the driver	Cut out of ACM/LCM/CCM/MCM shall be available to the driver
2F_05.01.ISSUES.-.Zefiro-Europe.25	[2F_05.01.Zefiro-Europe.CONCEPT.213]: [2F_05.01.Zefiro-Europe.CONCEPT.214]: [2F_05.01.Zefiro-Europe.CONCEPT.628]:	Traction test enabled in Low voltage	Traction test shall be enabled also if HV is not enabled
2F_05.01.ISSUES.-.Zefiro-Europe.26	[2F_05.01.Zefiro-Europe.CONCEPT.657]: [2F_05.01.Zefiro-Europe.CONCEPT.669]:	MCF start logic improvement	Start logic of MCF improvements after faulty conditions verified during commercial service
2F_05.01.ISSUES.-.Zefiro-Europe.27	[2F_05.01.Zefiro-Europe.CONCEPT.255]: [2F_05.01.Zefiro-Europe.CONCEPT.690]:	CBM test	New requirement
2F_05.01.ISSUES.-.Zefiro-Europe.31	[2F_05.01.Zefiro-Europe.CONCEPT.669]:	Re-start after permanent fault of the motor fan	New requirement
2F_05.01.ISSUES.-.Zefiro-Europe.42	[2F_05.01.Zefiro-Europe.CONCEPT.655]:	After missing 400 V permanent fault is reset	Change of the permanent fault

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2F_05.01.ISSUES.-.Zefiro-Europe.43	[2F_05.01.Zefiro-Europe.CONCEPT.669]:	After 3 attempts the TCMF is blocked until permanent fault is set to zero	Change of blocked motor fan after 3 re-starts
2F_05.01.ISSUES.-.Zefiro-Europe.49	[2F_05.01.Zefiro-Europe.CONCEPT.655]: [2F_05.01.Zefiro-Europe.CONCEPT.656]:	After missing 400 V permanent fault is reset	Change of the permanent fault
2F_05.01.ISSUES.-.Zefiro-Europe.51	[[2F_05.01.Zefiro-Europe.CONCEPT.669]:	After 3 attempts the TCMF is blocked until permanent fault is set to zero	Change of blocked motor fan after 3 re-starts

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